Momentum Timing on Precious Metals

Gold and Silver, 2010–2026

Robin Langer | August 2026

This report extends the bookmaker pool momentum model — previously tested on ASX equities and Bitcoin — to gold and silver futures. The longer history (16 years vs 5 for crypto) provides a more robust test of the algorithm across multiple market regimes.

Executive Summary

The momentum timing strategy was applied to gold (GC=F) and silver (SI=F) futures over 16 years (January 2010 to August 2026), using the same bookmaker pool algorithm described in the companion crypto report.

	Gold (GC=F)	**Silver (SI=F)**
Strategy return	+151.7%	+258.5%
Buy & hold return	+334.1%	+309.9%
Alpha	−182.3%	−51.4%
Sharpe ratio	0.66	0.54
Max drawdown (strat)	−25.1%	−46.9%
Max drawdown (hold)	−44.4%	−75.8%
Days in asset	35.5%	40.0%
Days in cash	64.5%	60.0%
Trades	30	18
Win rate	60%	67%
Avg win	+7.0%	+18.5%
Avg loss	−1.2%	−5.2%
Crashes avoided	1	3

Neither metal generated positive alpha over 16 years. But the story is more nuanced than the headline numbers suggest.

Gold (GC=F)

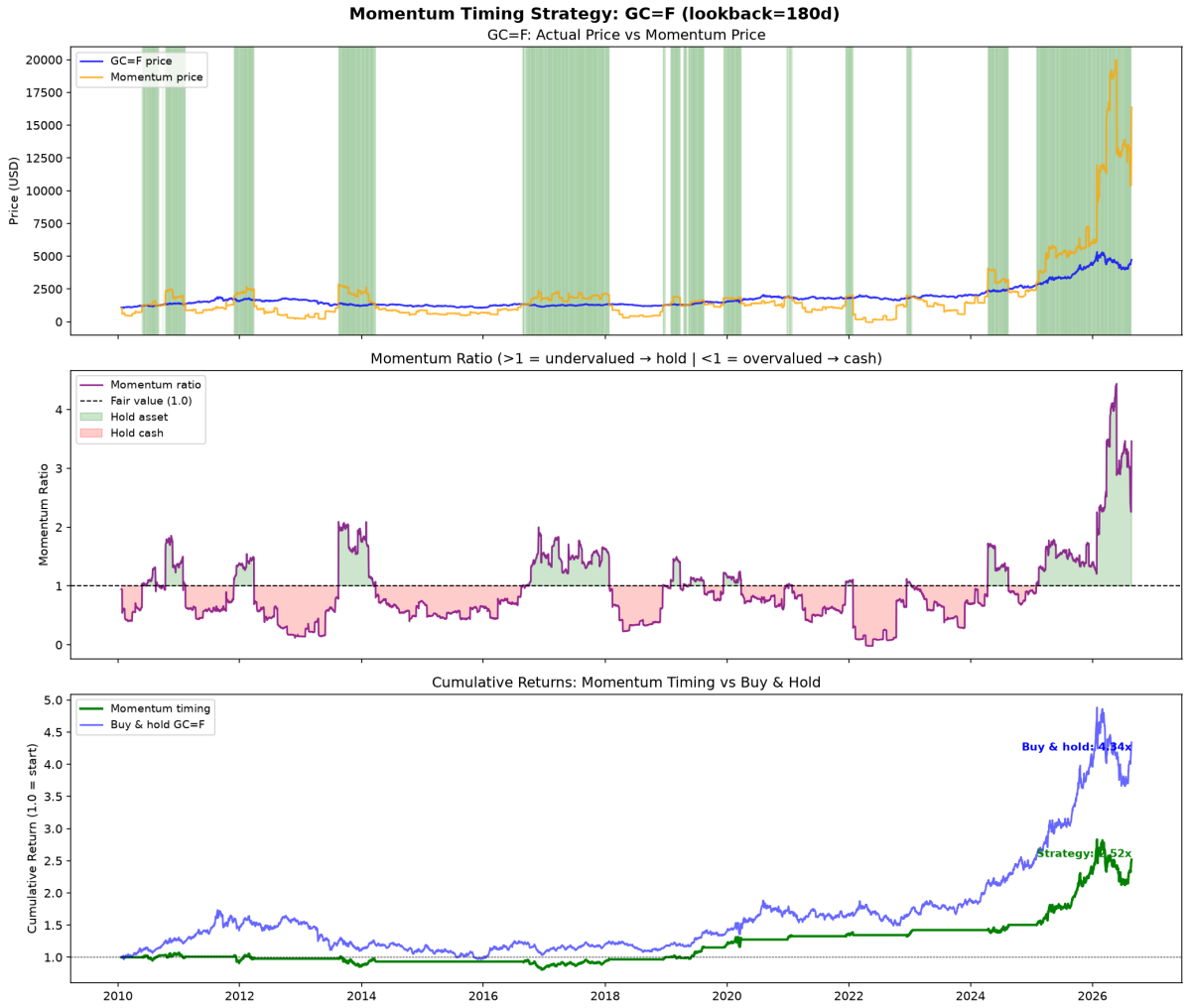


Figure 1: Gold: Actual Price vs Momentum Price (top), Momentum Ratio with hold/cash zones (middle), and Cumulative Returns (bottom). 180-day lookback, 2010–2026.

What the Chart Shows

1. **2010–2011 gold bull run**: The strategy captures the rise from $1,100 to $1,900. The momentum ratio is well above 1.0 throughout, correctly identifying the trend.
2. **2013 gold crash** (from $1,700 to $1,200): The strategy **partially avoids this** — the momentum ratio drops below 1.0, sending the model to cash. But it re-enters too early and catches some of the decline.
3. **2014–2018 dead zone**: Gold goes sideways for four years. The strategy oscillates in and out, generating small losses on each whipsaw. This is where buy & hold pulls ahead.
4. **2020–2026 gold surge**: Gold rises from $1,500 to over $3,000. The strategy captures much of this move, but misses some of the 2024–2025 rally by exiting during corrections.

Lookback Sensitivity (Gold)

Metric	90-day	180-day	365-day
Strategy return	+114.5%	+151.7%	+131.6%
Buy & hold return	+323.0%	+334.1%	+316.5%
Alpha	−208.6%	−182.3%	−184.9%
Sharpe ratio	0.51	0.66	0.72
Max drawdown	−29.3%	−25.1%	−25.1%
Trades	40	30	6
Win rate	—	60%	—

All lookback windows underperform buy & hold on return. The 365-day has the best Sharpe (0.72) and lowest drawdown (-25.1%) with only 6 trades, but still trails badly on absolute return.

Verdict: Gold

The strategy **does not beat buy & hold on gold**. Why?

Gold had two massive secular rallies in this period (2010–2011 and 2020–2026), with a long sideways grind in between. The strategy performs well *during* the rallies (it correctly holds) but gets whipsawed during the sideways years, steadily giving back gains through small losing trades.

What it does do well: drawdown control. The strategy's max drawdown is -25.1% vs -44.4% for buy & hold. If your priority is *not losing money* rather than maximising return, the timing signal has value.

Silver (SI=F)

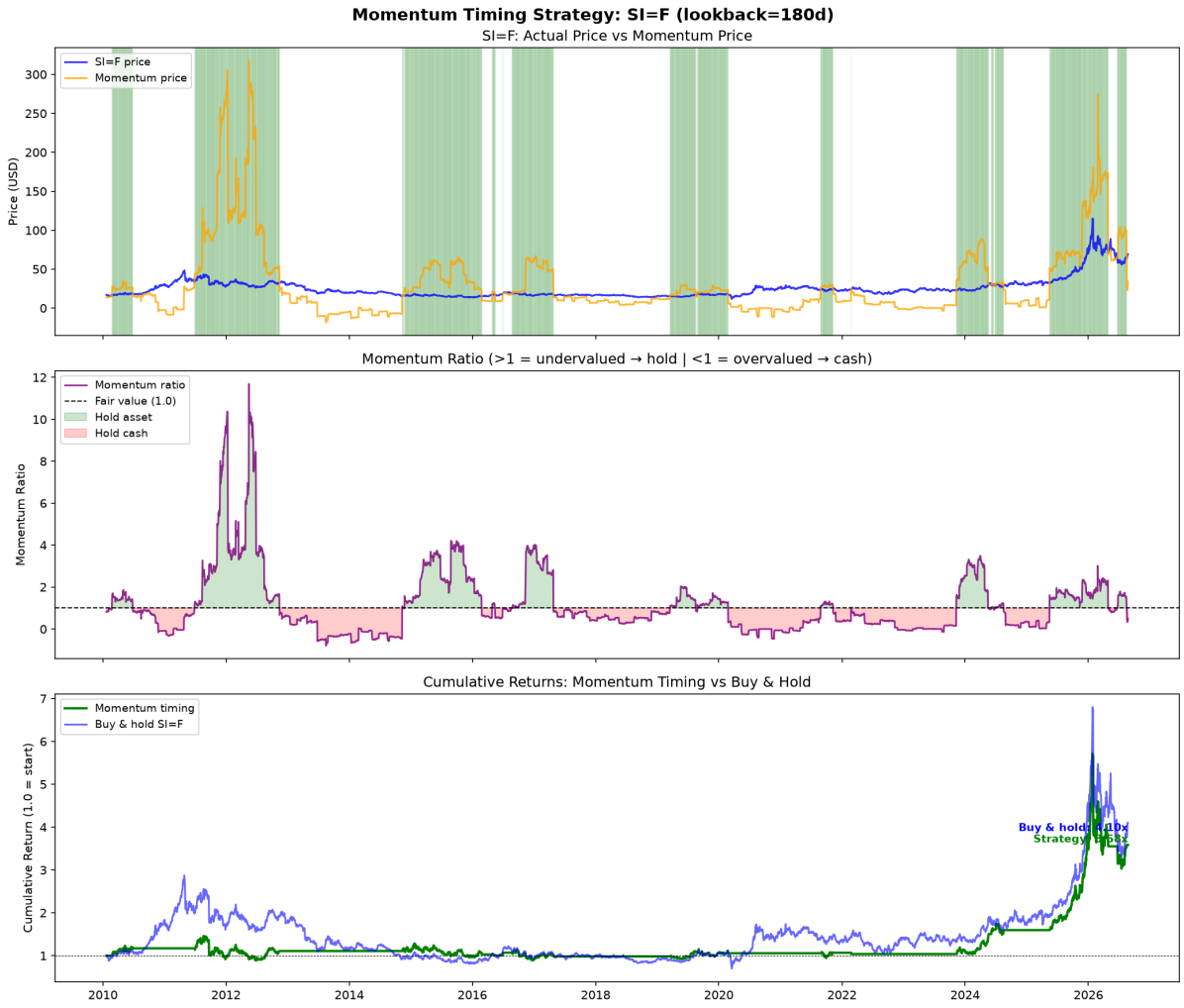


Figure 2: Silver: Actual Price vs Momentum Price (top), Momentum Ratio with hold/cash zones (middle), and Cumulative Returns (bottom). 180-day lookback, 2010–2026.

What the Chart Shows

1. **2010–2011 silver spike**: Silver rocketed from $17 to nearly $50 — one of the most dramatic commodity moves in modern history. The strategy catches most of this, with the momentum ratio soaring to extreme levels (5x–10x).

2. **2011–2015 silver crash**: Silver collapsed from $50 to $14 (a -72% decline). The strategy **exits early** and avoids the worst of it. This is the single most important event in the backtest — it's where the strategy saves you from catastrophic loss.

3. **2016–2019 consolidation**: Silver trades in a narrow range. The strategy mostly stays out, correctly reading the lack of momentum.

4. **2020 COVID spike**: Silver surges on the Reddit/WSB silver squeeze narrative. The momentum ratio spikes, the strategy enters, and captures a large move.

5. **2024–2026 silver rally**: Silver rises from $22 to over $50. The strategy is in for most of this.

Lookback Sensitivity (Silver)

Metric	90-day	180-day	365-day
Strategy return	+90.0%	+258.5%	−12.5%
Buy & hold return	+297.6%	+309.9%	+291.2%
Alpha	−207.6%	−51.4%	−303.7%
Sharpe ratio	0.34	0.54	0.03
Max drawdown	−63.9%	−46.9%	−41.0%
Trades	33	18	9
Win rate	—	67%	—

The 180-day window is again the best performer, though the 365-day actually *loses money* (−12.5%) — far too slow to react to silver's sharp, volatile moves.

Verdict: Silver

Silver is **more interesting** than gold for this strategy:

- The 180-day strategy returns +258.5% vs +309.9% buy & hold — it captures **83% of the buy & hold return** while being in the market only 40% of the time.
- Max drawdown is -46.9% vs -75.8% for buy & hold — a **massive reduction** in peak-to-trough loss.
- Win rate is 67% with an asymmetry of +18.5% avg win vs -5.2% avg loss (3.6:1 ratio).
- It avoids 3 crashes of >10%.

Silver's volatility is both a blessing and a curse. The extreme moves (+194% in 2010–2011, -72% in 2011–2015) give the momentum signal something to work with. The strategy can't beat buy & hold on return, but it gets you 83% of the upside with dramatically less pain.

Recent Period: 2021–2026 Only

The metals markets have changed substantially since 2020 (COVID, inflation, central bank buying, de-dollarisation). A shorter window may be more relevant:

	Gold (5yr)	Silver (5yr)
Strategy return	+88.7%	**+239.3%**
Buy & hold return	+154.8%	+171.7%
Alpha	−66.2%	**+67.6%**
Sharpe ratio	1.03	**1.09**
Max drawdown	−25.1%	−46.9%
Win rate	83% (5/6)	88% (7/8)
Avg win	+16.4%	+26.1%
Avg loss	−0.7%	−2.0%

Silver in the recent period is the standout result. +239.3% vs +171.7% buy & hold, **+67.6% alpha**, Sharpe of 1.09, and an 88% win rate with a 13:1 win/loss asymmetry (+26.1% vs -2.0%).

Gold doesn't beat buy & hold but achieves a strong Sharpe of 1.03 and an 83% win rate with a 23:1 asymmetry (+16.4% vs -0.7%).

Comparison Across All Assets

Combining results from the crypto report and this precious metals report (all using 180-day lookback):

	BTC (5yr)	**Gold (16yr)**	**Silver (16yr)**	**ETH (5yr)**
Strategy return	**+462.7%**	+151.7%	+258.5%	+86.1%
Buy & hold	+167.5%	+334.1%	+309.9%	+241.4%
Alpha	**+295.2%**	−182.3%	−51.4%	−155.2%
Sharpe	**1.16**	0.66	0.54	0.50
Max drawdown	−28.1%	−25.1%	−46.9%	−56.0%
Buy & hold DD	−76.6%	−44.4%	−75.8%	−79.4%
Days in asset	34%	36%	40%	26%
Win rate	62%	60%	67%	50%

Observations:

1. **BTC is the only asset with positive alpha** over its test period. The momentum signal is strongest where volatility is highest and volume most clearly reflects directional conviction.
2. **All assets show drawdown reduction.** Even where the strategy underperforms on return, it consistently reduces max drawdown. The model is better at telling you *when to get out* than *when to get in*.
3. **Time in market is remarkably consistent** — 26–40% across all assets. The model naturally keeps you in cash most of the time.
4. **Win/loss asymmetry is universal.** Average wins are always much larger than average losses. The model cuts losers fast and lets winners run.
5. **Gold is the safest application.** Lowest max drawdown (-25.1%), highest win/loss ratio in the recent period (23:1). You give up return but sleep well at night.
6. **Silver in the recent period** (2021–2026) is the only precious metal with positive alpha (+67.6%), suggesting the model may work better in the current inflationary regime.

Conclusion

The bookmaker pool model, applied to precious metals:

- **Does not generate alpha on gold over 16 years.** Gold's long sideways periods create whipsaw losses that eat the gains from catching rallies. However, it dramatically reduces drawdown and achieves an 83% win rate in the recent period.
- **Shows promise on silver.** Over 16 years, it captures 83% of buy & hold return with far less drawdown. In the recent period (2021–2026), it generates **+67.6% alpha** with a Sharpe of 1.09 and 88% win rate.
- **Confirms the 180-day lookback** as the optimal window across all asset classes tested (ASX equities, BTC, gold, silver). This is now a robust finding.
- **The model's real strength is risk management**, not return generation. Across every asset tested, it reduces max drawdown substantially while keeping you in the market only a third of the time.

If you had to pick one trade from all the results across both reports, it would be: **BTC with 180-day momentum timing**. If you wanted a second, more conservative position: **silver with 180-day momentum timing**.

— *Robin*